

ANSWERS

1.

- (a) field strength = potential gradient M1
correct sign OR directions discussed A1
- (b) area is $21.2 \text{ cm}^2 \pm 0.4 \text{ cm}^2$ C2
(if outside $\pm 0.4 \text{ cm}^2$ but within $\pm 0.8 \text{ cm}^2$, allow 1 mark)
 1.0 cm^2 represents $(1.0 \times 10^{-2} \times 2.5 \times 10^3 =) 25 \text{ V}$ C1
potential difference = 530 V A1
- (c) $\frac{1}{2}mv^2 = qV$
 $\frac{1}{2} \times 9.1 \times 10^{-31} \times v^2 = 1.6 \times 10^{-19} \times 530$ C1
 $v = 1.37 \times 10^7 \text{ ms}^{-1}$ A1
- (d) (i) $d = 0$ B1
- (ii) acceleration decreases then increases B1
some quantitative analysis (e.g. minimum at 4.0 cm) B1
(any suggestion that acceleration becomes zero or that there is a deceleration scores 0/2)

2.

- (a) (i) either lines directed away from sphere
or lines go from positive to negative
or line shows direction of force on positive charge M1
so positively charged A1
- (ii) either all lines (appear to) radiate from centre
or all lines are normal to surface of sphere B1
- (b) tangent to curve B1
in correct position and direction B1
- (c) (i) $V = (0.76 \times 10^{-9}) / (4\pi \times 8.85 \times 10^{-12} \times 0.024)$ C1
 $= 285 \text{ V}$ A1
- (ii) negative charge is induced on (inside of) box M1
formula applies to isolated (point) charge
OR less work done moving test charge from infinity A1
so potential is lower A1
- (d) either gravitational field is always attractive
or field lines must be directed towards both box and sphere B1

3.

(a) work done moving unit positive charge from infinity to the point M1
A1

(b) (i) $x = 18 \text{ cm}$ A1

(ii) $V_A + V_B = 0$ C1

$$(3.6 \times 10^{-9}) / (4\pi\epsilon_0 \times 18 \times 10^{-2}) + q / (4\pi\epsilon_0 \times 12 \times 10^{-2}) = 0$$

$$q = -2.4 \times 10^{-9} \text{ C}$$

(use of $V_A = V_B$ giving $2.4 \times 10^{-9} \text{ C}$ scores one mark)

(c) field strength = (–) gradient of graph B1

force = charge $\times$ gradient / field strength or force $\propto$ gradient B1

force largest at $x = 27 \text{ cm}$ B1

4.

(a) work done per / on unit positive chargeM1
moving charge from infinity to the pointA1

(b) (i) α -particle and gold nucleus repel each otherB1

all kinetic energy of α -particle converted into electric potential energyB1

(ii) 1 potential energy = $(79 \times 2 \times \{1.6 \times 10^{-19}\}^2) / (4\pi \times 8.85 \times 10^{-12} \times d)$ C1

kinetic energy = $4.8 \times 1.6 \times 10^{-13} = 7.68 \times 10^{-13} \text{ J}$ C1

equating to give $d = 4.7 \times 10^{-14} \text{ m}$ A1

(ii) 2 $F = Qq / 4\pi\epsilon_0 d \times 1 / d = 7.68 \times 10^{-13} \times 1 / (4.7 \times 10^{-14})$ C1

= 16 NA1

5.

(a) ability to do work B1
as a result of the position/shape, etc. of an object B1

(b) (i) 1 $\Delta E_{\text{gpe}} = GMm/r$ C1
 $= (6.67 \times 10^{-11} \times \{2 \times 1.66 \times 10^{-27}\}^2) / (3.8 \times 10^{-15})$ C1
 $= 1.93 \times 10^{-49} \text{ J}$ A1

2 $\Delta E_{\text{epe}} = Qq / 4\pi\epsilon_0 r$ C1
 $= (1.6 \times 10^{-19})^2 / (4\pi \times 8.85 \times 10^{-12} \times 3.8 \times 10^{-15})$ C1
 $= 6.06 \times 10^{-14} \text{ J}$ A1

(ii) idea that $2E_K = \Delta E_{\text{epe}} - \Delta E_{\text{gpe}}$ B1
 $E_K = 3.03 \times 10^{-14} \text{ J}$
 $= (3.03 \times 10^{-14}) / 1.6 \times 10^{-13}$ M1
 $= 0.19 \text{ MeV}$ A0

(iii) fusion may occur / may break into sub-nuclear particles B1

6.

(a) force $= q_1 q_2 / 4\pi\epsilon_0 x^2$ C1
 $= (6.4 \times 10^{-19})^2 / (4\pi \times 8.85 \times 10^{-12} \times \{12 \times 10^{-6}\}^2)$ C1
 $= 2.56 \times 10^{-17} \text{ N}$ A1

(b) potential at P is same as potential at Q B1
work done $= q\Delta V$ M1
 $\Delta V = 0$ so zero work done A0

(c) at midpoint, potential is $2 \times (6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 6 \times 10^{-6})$ C1
at P, potential is $(6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 3 \times 10^{-6}) + (6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 9 \times 10^{-6})$ C1
change in potential $= (6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 9 \times 10^{-6})$
energy $= 1.6 \times 10^{-19} \times (6.4 \times 10^{-19}) / (4\pi\epsilon_0 \times 9 \times 10^{-6})$ C1
 $= 1.0 \times 10^{-22} \text{ J}$ A1

7.

(a) work done in bringing unit positive charge M1
from infinity (to that point) A1

(b) (i) field strength is potential gradient B1

(ii) field strength proportional to force (on particle Q) B1
potential gradient proportional to gradient of (potential energy) graph B1
so force is proportional to the gradient of the graph A0

(c) energy = $5.1 \times 1.6 \times 10^{-19}$ (J) C1
 potential energy = $Q_1 Q_2 / 4\pi\epsilon_0 r$ C1
 $5.1 \times 1.6 \times 10^{-19} = (1.6 \times 10^{-19})^2 / 4\pi \times 8.85 \times 10^{-12} \times r$ C1
 $r = 2.8 \times 10^{-10}$ m A1

(d) (i) work is got out as x decreases M1
 so opposite sign A1
 (ii) energy would be doubled B1
 gradient would be increased B1

8.

(a) region (of space) where a particle / body experiences a force B1

(b) similarity: e.g. force $\propto 1 / r^2$
 potential $\propto 1 / r$ B1

difference: e.g. gravitation force (always) attractive B1
 electric force attractive or repulsive B1

(c) either ratio is $Q_1 Q_2 / 4\pi\epsilon_0 m_1 m_2 G$ C1
 $= (1.6 \times 10^{-19})^2 / 4\pi \times 8.85 \times 10^{-12} \times (1.67 \times 10^{-27})^2 \times 6.67 \times 10^{-11}$ C1
 $= 1.2 \times 10^{36}$ A1
 or $F_E = 2.30 \times 10^{-28} \times R^{-2}$ (C1)
 $F_G = 1.86 \times 10^{-64} \times R^{-2}$ (C1)
 $F_E / F_G = 1.2 \times 10^{36}$ (A1)

9.

(a) (i) zero field (strength) inside spheres B1

(ii) either field strength is zero
 or the fields are in opposite directions M1
 at a point between the spheres A1

(b) (i) field strength is (–) potential gradient (not V/x) B1

(ii) 1. field strength has maximum value B1
 at $x = 11.4$ cm B1

2. field strength is zero B1
 either at $x = 7.9$ cm (allow ± 0.3 cm)
 or at 0 to 1.4 cm or 11.4 cm to 12 cm B1

10.

(a) force per unit positive charge acting on a stationary charge B1

(b) (i) $E = Q / 4\pi\epsilon_0 r^2$ C1

$$Q = 1.8 \times 10^4 \times 10^2 \times 4\pi \times 8.85 \times 10^{-12} \times (25 \times 10^{-2})^2$$

M1

$$Q = 1.25 \times 10^{-5} \text{ C} = 12.5 \mu\text{C}$$

A0

(ii) $V = Q / 4\pi\epsilon_0 r$

$$= (1.25 \times 10^{-5}) / (4\pi \times 8.85 \times 10^{-12} \times 25 \times 10^{-2})$$

C1

$$= 4.5 \times 10^5 \text{ V}$$

A1

(Do not allow use of $V = Er$ unless explained)

11.

(a) (i) as r decreases, energy decreases/work got out (due to attraction) so point mass is negatively charged M1
A1

(ii) electric potential energy = charge $\times$ electric potential B1

electric field strength is potential gradient B1

field strength = gradient of potential energy graph/charge A0

(b) tangent drawn at (4.0, 14.5) B1

$$\text{gradient} = 3.6 \times 10^{-24}$$

A2

(for $< \pm 0.3$ allow 2 marks, for $< \pm 0.6$ allow 1 mark)

$$\text{field strength} = (3.6 \times 10^{-24}) / (1.6 \times 10^{-19})$$

$$= 2.3 \times 10^{-5} \text{ V m}^{-1} \text{ (allow ecf from gradient value)} \quad \text{A1}$$

(one point solution for gradient leading to $2.3 \times 10^{-5} \text{ V m}^{-1}$ scores 1 mark only)

12.

(a) (i) (tangent to line gives) direction of force on a (small test) mass B1

(ii) (tangent to line gives) direction of force on a (small test) charge M1

charge is positive A1

- (b) similarity:
 e.g. radial fields
 lines normal to surface
 greater separation of lines with increased distance from sphere
 field strength $\propto 1 / (\text{distance to centre of sphere})^2$
 (allow any sensible answer) B1
- difference:
 e.g. gravitational force (always) towards sphere B1
 electric force direction depends on sign of charge on sphere / towards or away from sphere B1
 e.g. gravitational field/force is attractive (B1)
 electric field/force is attractive or repulsive (B1)
 (allow any sensible comparison)
- (c) gravitational force = $1.67 \times 10^{-27} \times 9.81$
 $= 1.6 \times 10^{-26} \text{ N}$ A1
 electric force = $1.6 \times 10^{-19} \times 270 / (1.8 \times 10^{-2})$ C1
 $= 2.4 \times 10^{-15} \text{ N}$ A1
 electric force very much greater than gravitational force B1
- 13.
- (a) work done moving unit positive charge from infinity (to the point) M1
 A1
- (b) (gain in) kinetic energy = change in potential energy B1
 $\frac{1}{2}mv^2 = qV$ leading to $v = (2Vq/m)^{1/2}$ B1
- (c) either $(2.5 \times 10^5)^2 = 2 \times V \times 9.58 \times 10^7$ C1
 $V = 330 \text{ V}$ M1
 this is less than 470 V and so 'no' A1
- or $v = (2 \times 470 \times 9.58 \times 10^7)$ (C1)
 $v = 3.0 \times 10^5 \text{ ms}^{-1}$ (M1)
 this is greater than $2.5 \times 10^5 \text{ ms}^{-1}$ and so 'no' (A1)
- or $(2.5 \times 10^5)^2 = 2 \times 470 \times (q/m)$ (C1)
 $(q/m) = 6.6 \times 10^7 \text{ C kg}^{-1}$ (M1)
 this is less than $9.58 \times 10^7 \text{ C kg}^{-1}$ and so 'no' (A1)

14.

- (a) (i) force proportional to product of (two) charges and inversely proportional to square of separation
reference to point charges M1
A1
- (ii) $F = 2 \times (1.6 \times 10^{-19})^2 / \{4\pi \times 8.85 \times 10^{-12} \times (20 \times 10^{-6})^2\}$ C1
 $F = 1.15 \times 10^{-18} \text{ N}$ A1
- (b) (i) force per unit charge M1
on *either* a stationary charge
or a positive charge A1
- (ii) 1. electric field is a vector quantity
electric fields are in opposite directions
charges repel
Any two of the above, 1 each B2
2. graph: line always between given lines M1
crosses x-axis between 11.0 μm and 12.3 μm A1
reasonable shape for curve A1

15.

- (a) work done bringing unit positive charge from infinity (to the point) M1
A1
- (b) (i) *either* both potentials are positive/same sign M1
so same sign A1
or gradients are positive & negative (so fields in opposite directions) (M1)
so same sign (A1)
- (ii) the individual potentials are summed B1
- (iii) allow value of x between 10 nm and 13 nm A1
- (iv) $V = 0.43 \text{ V}$ (allow $0.42 \text{ V} \rightarrow 0.44 \text{ V}$) M1
energy $= 2 \times 1.6 \times 10^{-19} \times 0.43$ A1
 $= 1.4 \times 10^{-19} \text{ J}$ A1

16.

- (a) graph: straight line at constant potential = V_0 from $x = 0$ to $x = r$ B1
 curve with decreasing gradient M1
 passing through $(2r, 0.50V_0)$ and $(4r, 0.25V_0)$ A1
- (b) graph: straight line at $E = 0$ from $x = 0$ to $x = r$ B1
 curve with decreasing gradient from (r, E_0) M1
 passing through $(2r, \frac{1}{4}E_0)$ A1
(for 3rd mark line must be drawn to $x = 4r$ and must not touch x-axis)

17.

- (a) (i) $F_E = Q_1Q_2/4\pi\epsilon_0r^2$ C1
 $= 8.99 \times 10^9 \times (1.6 \times 10^{-19})^2 / (2.0 \times 10^{-15})^2$
 $= 58 \text{ N}$ A1
- (ii) $F_G = Gm_1m_2/r^2$ C1
 $= 6.67 \times 10^{-11} \times (1.67 \times 10^{-27})^2 / (2.0 \times 10^{-15})^2$
 $= 4.7 \times 10^{-35} \text{ N}$ A1
- (b) (i) force of repulsion (much) greater than force of attraction B1
 must be some other force of attraction M1
 to hold nucleus together A1
- (Do not allow if $F_G > F_E$ in (a) or one of the forces not calculated in (a))*
- (ii) outside nucleus there is repulsion between protons B1
either attractive force must act only in nucleus
or if not short range, all nuclei would stick together B1

18.

- (a) work done in moving unit positive charge M1
 from infinity (to the point) A1
- (b) (i) inside the sphere, the potential would be constant B1
- (ii) for point charge, V_x is constant B1
 co-ordinates clear and determines two values of V_x at least 4 cm apart M1
 conclusion made clear A1
- (c) $q = 4\pi\epsilon_0 Vx$
 $q = 4\pi \times 8.85 \times 10^{-12} \times 180 \times 1.0 \times 10^{-2}$ M1
 $= 2.0 \times 10^{-10} \text{ C}$ A1

19.

(a) work done/energy in moving unit positive charge from infinity (to the point) M1
A1

(b) (i) $V = q/4\pi\epsilon_0 r$
at 16 kV, $q = 3.0 \times 10^{-8} \text{ C}$

$$r = (3.0 \times 10^{-8}) / (4\pi \times 8.85 \times 10^{-12} \times 16 \times 10^3) \quad \text{C1}$$

$$= 1.69 \times 10^{-2} \text{ m (allow 2 s.f.)} \quad \text{A1}$$

(allow any answer which rounds to 1.7×10^{-2})

(ii) energy is/represented by area 'below' line C1
energy = $\frac{1}{2}qV$
 $= \frac{1}{2} \times 24 \times 10^3 \times 4.5 \times 10^{-8}$ C1
 $= 5.4 \times 10^{-4} \text{ J}$ A1

(c) $V = q/4\pi\epsilon_0 r$ and $E = q/4\pi\epsilon_0 r^2$ giving $Er = V$ B1
 $2.0 \times 10^6 \times 1.7 \times 10^{-2} = V$ C1
 $V = 3.4 \times 10^4 \text{ V}$ A1

20.

(a) (magnitude of electric field strength is the potential gradient B1
use of gradient at $x = 4.0 \text{ cm}$ M1
gradient = $4.5 \times 10^4 \text{ NC}^{-1}$ (allow $\pm 0.3 \times 10^4$) A1

or

$$V = \frac{Q}{4\pi\epsilon_0 x} \text{ and } E = \frac{Q}{4\pi\epsilon_0 x^2} \text{ leading to } E = \frac{V}{x} \quad \text{(B1)}$$

$$E = 1.8 \times 10^3 / 0.04 \quad \text{(M1)}$$

$$= 4.5 \times 10^4 \text{ NC}^{-1} \quad \text{(A1)}$$

(b) (i) $3.6 \times 10^3 \text{ V}$ A1

(ii) capacitance = Q/V C1
 $= (8.0 \times 10^{-9}) / (3.6 \times 10^3)$
 $= 2.2 \times 10^{-12} \text{ F}$ A1

21.

- (a) work done bringing/moving per unit positive charge from infinity (to the point) M1
A1
- (b) (i) slope/gradient (of the line/graph/tangent) B1
(allow dV/dx , but **not** $\Delta V/\Delta x$ or V/x)
(allow potential gradient)
(negative sign not required)
- (ii) maximum at surface of sphere A or at $x = 0$ (cm) B1
zero at $x = 6$ (cm) B1
then increases but in opposite direction B1
(any mention of attraction max. 2/3)
- (c) (i) M shown between $x = 5.5$ cm and $x = 6.5$ cm B1
- (ii) 1. $\Delta V = (570 - 230) = 340$ V (allow 330 V to 340 V) A1
2. $q(\Delta)V = \frac{1}{2}mv^2$ or change/loss in PE = change/gain in KE or $\Delta E_K = \Delta E_P$ B1
- $4.8 \times 10^7 \times 340 = \frac{1}{2}v^2$ C1
 $v^2 = 3.26 \times 10^{10}$
 $v = 1.8 \times 10^5 \text{ m s}^{-1}$ (not 1 s.f.) A1

22.

- (a) (i) force proportional to the product of the two/point charges B1
and inversely proportional to the square of their separation B1
- (ii) 1. force radially away from sphere/to right/to east B1
2. (maximum) at/on surface of sphere or $x = r$ B1
3. $F \propto 1/x^2$ or $F = q_1q_2/(4\pi\epsilon_0x^2)$ C1
- ratio = 16 A1
- (b) $E = q/(4\pi\epsilon_0x^2)$ or $E \propto q$ C1
- maximum charge = $(2.0/1.5) \times 6.0 \times 10^{-7}$ C1
= 8.0×10^{-7} C
- additional charge = 2.0×10^{-7} C A1

23.

(a) charges in metal do not move B1
 no (resultant) force on charges so no (electric) field B1
 (allow 1/2 for "no field inside sphere")

(b) either average field strength = $\frac{1}{2} (28 + 54) \text{ NC}^{-1}$ C1

average force = $8.5 \times 10^{-9} \times \frac{1}{2} (28 + 54)$ C1
 = $3.49 \times 10^{-7} \text{ N}$

change in potential energy = $3.49 \times 10^{-7} \times 2.0 \times 10^{-2}$
 = $7.0 \times 10^{-9} \text{ J}$ (allow 1 s.f.) A1

(allow range 54 ± 1)

or (for a point charge) $V = Ex$ (C1)

$\Delta V = (54 \times 5.0 \times 10^{-2}) - (28 \times 7.0 \times 10^{-2})$ (C1)

change in potential energy = $8.5 \times 10^{-9} \times (2.70 - 1.96)$
 = $6.3 \times 10^{-9} \text{ J}$ (allow 1 s.f.) (A1)

(allow range 54 ± 1)

or ΔV is area under curve (C1)

$\Delta V = 0.74 \text{ V}$ (C1)

change in potential energy = $8.5 \times 10^{-9} \times 0.74$
 = $6.3 \times 10^{-9} \text{ J}$ (allow 1 s.f.) (A1)

(allow range 0.70 to 0.84)

24.

(a) sketch: from $x = 0$ to $x = R$, potential is constant at V_S B1

smooth curve through (R, V_S) and $(2R, 0.5V_S)$ B1

smooth curve continues to $(3R, 0.33V_S)$ B1

(b) sketch: from $x = 0$ to $x = R$, field strength is zero B1

smooth curve through (R, E) and $(2R, 0.25E)$ B1

smooth curve continues to $(3R, 0.11E)$ B1

25.

(a) lines perpendicular to surface

or

lines are radial

M1

lines appear to come from centre

A1

(b) (i) $F_E = (1.6 \times 10^{-19})^2 / 4\pi\epsilon_0 x^2$

C1

$$F_G = G \times (1.67 \times 10^{-27})^2 / x^2$$

C1

$$F_E / F_G = (1.6 \times 10^{-19})^2 \times (8.99 \times 10^9) / [(1.67 \times 10^{-27})^2 \times (6.67 \times 10^{-11})]$$

$$= 1.2 (1.24) \times 10^{36}$$

A1

(ii) $F_E \gg F_G$

B1

26.

(a) in an electric field, charges (in a conductor) would move

B1

no movement of charge so zero field strength

or

charge moves until $F = 0$ / $E = 0$

B1

or

charges in metal do not move

(B1)

no (resultant) force on charges so no (electric) field

(B1)

(b) at P, $E_A = (3.0 \times 10^{-12}) / [4\pi\epsilon_0(5.0 \times 10^{-2})^2]$ (= 10.79 NC⁻¹)

M1

at P, $E_B = (12 \times 10^{-12}) / [4\pi\epsilon_0(10 \times 10^{-2})^2]$ (= 10.79 NC⁻¹)

M1

or

$$(3.0 \times 10^{-12}) / [4\pi\epsilon_0(5.0 \times 10^{-2})^2] - (12 \times 10^{-12}) / [4\pi\epsilon_0(10 \times 10^{-2})^2] = 0$$

or

$$(3.0 \times 10^{-12}) / [4\pi\epsilon_0(5.0 \times 10^{-2})^2] = (12 \times 10^{-12}) / [4\pi\epsilon_0(10 \times 10^{-2})^2]$$

(M2)

fields due to charged spheres are (equal and) opposite in direction, so $E = 0$

A1

(c) potential = $8.99 \times 10^9 \{ (3.0 \times 10^{-12}) / (5.0 \times 10^{-2}) + (12 \times 10^{-12}) / (10 \times 10^{-2}) \}$

C1

$$= 1.62 \text{ V}$$

A1

(d) $\frac{1}{2}mv^2 = qV$

$$E_K = \frac{1}{2} \times 107 \times 1.66 \times 10^{-27} \times v^2 \quad \text{C1}$$

$$qV = 47 \times 1.60 \times 10^{-19} \times 1.62 \quad \text{C1}$$

$$v^2 = 1.37 \times 10^8$$

$$v = 1.2 \times 10^4 \text{ ms}^{-1} \quad \text{A1}$$

27.

(a) $E = 0$ or $E_A = (-)E_B$ (at $x = 11 \text{ cm}$) B1

$$Q_A/x^2 = Q_B/(20-x)^2 = 11^2/9^2 \quad \text{C1}$$

$$Q_A/Q_B \text{ or ratio} = 1.5 \quad \text{A1}$$

or

$$E \propto Q \text{ because } r \text{ same or } E = Q/4\pi\epsilon_0 r^2 \text{ and } r \text{ same} \quad \text{(B1)}$$

$$Q_A/Q_B = 48/32 \quad \text{(C1)}$$

$$Q_A/Q_B \text{ or ratio} = 1.5 \quad \text{(A1)}$$

(b) (i) for max. speed, $\Delta V = (0.76 - 0.18) \text{ V}$ or $\Delta V = 0.58 \text{ V}$ C1

$$q\Delta V = \frac{1}{2}mv^2$$

$$2 \times (1.60 \times 10^{-19}) \times 0.58 = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2 \quad \text{C1}$$

$$v^2 = 5.59 \times 10^7$$

$$v = 7.5 \times 10^3 \text{ ms}^{-1} \quad \text{A1}$$

(ii) $\Delta V = 0.22 \text{ V}$ C1

$$2 \times (1.60 \times 10^{-19}) \times 0.22 = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2$$

$$v^2 = 2.12 \times 10^7$$

$$v = 4.6 \times 10^3 \text{ ms}^{-1} \quad \text{A1}$$

28.

5(a)	(loss in) kinetic energy of α -particle = $Qq / 4\pi\epsilon_0 r$ or $7.7 \times 10^{-13} = Qq / 4\pi\epsilon_0 r$	C1
	$7.7 \times 10^{-13} = 8.99 \times 10^9 \times 79 \times 2 \times (1.60 \times 10^{-19})^2 / r$	M1
	$r = 4.7 \times 10^{-14} \text{ m}$	A1
	r is closest distance of approach so radius less than this	
5(b)	force = $Qq / 4\pi\epsilon_0 r^2 = 4u \times a$	C1
	$8.99 \times 10^9 \times 79 \times 2 \times (1.60 \times 10^{-19})^2 / (4.7 \times 10^{-14})^2 = 4 \times 1.66 \times 10^{-27} \times a$	C1
	$a = 2.5 \times 10^{27} \text{ m s}^{-2}$	A1
5(c)	so that single interactions between nucleus and α -particle can be studied or so that multiple deflections with nucleus do not occur	B1

29.

6(a)	force proportional to product of charges and inversely proportional to the square of the separation	M1
	reference to point charges	A1
6(b)(i)	(near to each sphere,) fields are in opposite directions or point (between spheres) where fields are equal and opposite or point (between spheres) where field strength is zero	M1
	so same (sign of charge)	A1
6(b)(ii)	(at $x = 5.0 \text{ cm}$.) $E = 3.0 \times 10^3 \text{ V m}^{-1}$ and $a = qE / m$	C1
	$E = (1.60 \times 10^{-19} \times 3.0 \times 10^3) / (1.67 \times 10^{-27})$	C1
	$= 2.9 \times 10^{11} \text{ m s}^{-2}$	A1
6(c)	field strength or E is potential gradient or field strength is rate of change of (electric) potential	M1
	(field strength) maximum at $x = 6 \text{ cm}$	A1

30.

6(a)	electric field lines are radial/normal to surface (of sphere)	B1
	electric field lines <u>appear</u> to originate from centre (of sphere)	B1
6(b)(i)	tangent drawn at $x = 6.0 \text{ cm}$ and gradient calculation attempted	C1
	$E = 9.0 \times 10^4 \text{ NC}^{-1}$ (1 mark if in range ± 1.2 ; 2 marks if in range ± 0.6)	A2
	or	
	correct pair of values of V and x read from curved part of graph and substituted into $V = q / 4\pi\epsilon_0 x$	(C1)
	to give $q = 3.6 \times 10^{-8} \text{ C}$	(C1)
	(then $E = q / 4\pi\epsilon_0 x^2$ and $x = 6 \text{ cm}$ gives) $E = 9.0 \times 10^4 \text{ NC}^{-1}$	(A1)
	or	
	($E = q / 4\pi\epsilon_0 x^2$ and $V = q / 4\pi\epsilon_0 x$ and so) $E = V / x$	(C1)
	giving $E = 5.4 \times 10^3 / 0.060$	(C1)
	$= 9.0 \times 10^4 \text{ NC}^{-1}$	(A1)
6(b)(ii)	($R =$) 2.5 cm	B1
	potential inside a conductor is constant or field strength inside a conductor zero (so gradient is zero)	B1

31.

7(a)	work done per unit charge	B1
	(work done) moving positive charge from infinity (to the point)	B1
7(b)(i)	potential always same sign / potential is always positive so same sign of charge	B1
7(b)(ii)	1 from $x = 12 \text{ cm}$ to $x = 25 \text{ cm}$: speed increases and from $x = 27 \text{ cm}$ to $x = 31 \text{ cm}$: speed decreases	B1
	(from $x = 12 \text{ cm}$ to $x = 25 \text{ cm}$: speed increases) at decreasing rate or (from $x = 27 \text{ cm}$ to $x = 31 \text{ cm}$: speed decreases) at increasing rate	B1
	at $x = 26 \text{ cm}$: speed maximum	B1
	at 32 cm : speed still decreasing	B1
	2 $q \Delta V = \frac{1}{2}mv^2$ $3.2 \times 10^{-19} \times (2.14 - 1.43) \times 10^4 = \frac{1}{2} \times 6.6 \times 10^{-27} \times v^2$ $v^2 = 6.88 \times 10^{11}$	C1
	$v = 8.3 \times 10^5 \text{ m s}^{-1}$ (8.30)	A1

32.

6(a)	force per unit charge	B1
6(b)	$E = Q / (4\pi\epsilon_0 r^2)$	C1
	$2.0 \times 10^4 = Q / (4\pi \times 8.85 \times 10^{-12} \times 0.26^2)$	A1
	charge = $1.5 \times 10^{-7} \text{ C}$	
6(c)	charge (= $Q [52/26]^2$) = $4Q$	C1
	additional charge = $3Q$	A1

33.

6(a)(i)	work done per unit charge	B1
	work done moving positive charge from infinity (to the point)	B1
6(a)(ii)	field strength = potential gradient	M1
	'-' sign included or directions discussed	A1
6(b)(i)	gain in kinetic energy (= loss in potential energy) = charge $\times$ p.d. or $qV = \frac{1}{2}mv^2$	M1
	so v is independent of separation (because separation not in expressions)	A1
6(b)(ii)	(at $x = 0.40 \text{ cm}$), potential = $(-) 75 \times 0.40 / 1.2$ (= $(-) 25 \text{ V}$)	C1
	$\frac{1}{2}mv^2 = qV$	C1
	$\frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2 = 2 \times 1.60 \times 10^{-19} \times 25$	
	or	
	$a = Vq/dm$ and $v^2 = 2as$	(C1)
	$v^2 = (2 \times 75 \times 2 \times 1.60 \times 10^{-19} \times 0.40 \times 10^{-2}) / (1.2 \times 10^{-2} \times 4 \times 1.66 \times 10^{-27})$	(C1)
	$v = 4.9 \times 10^4 \text{ m s}^{-1}$	A1

34.

6(a)(i)	work done per unit charge	B1
	work done moving positive charge from infinity (to the point)	B1
6(a)(ii)	field strength = potential gradient	M1
	negative sign included or directions discussed	A1
6(b)	horizontal straight lines, at non-zero potential, within the spheres	B1
	magnitude of potential greater at surface of sphere A than at surface of sphere B	B1
	concave curve between A and B, with a minimum nearer to B	B1
	lines show V <u>positive</u> all the way from 0 to D	B1

35.

5(a)	region where charge experiences an (electric) force	B1
5(b)	graph: field strength zero from $x = 0$ to $x = R$	B1
	curve with negative gradient, decreasing from $x = R$ to $x = 3R$	B1
	line passes through field strength E at $x = R$,	B1
	line passes through field strength $0.25E$ at $x = 2R$ <u>and</u> field strength $0.11E$ at $x = 3R$	B1
5(c)	field strength = $q / 4\pi\epsilon_0 x^2$	C1
	$2.0 \times 10^6 = q / (4 \times \pi \times 8.85 \times 10^{-12} \times 0.26^2)$	C1
	$q = 1.5 \times 10^{-5} \text{ C}$	A1

36.

5(a)	force per unit charge	B1
	(force on) positive charge	B1
5(b)(i)	field changes <u>direction</u> (between A and B)/field is zero at a point (between A and B)	M1
	so charges have same sign	A1
5(b)(ii)	Any one from: - field is (also) influenced by charge B - charge A is not isolated/is not the only charge present - field is due to two/both charges - field is the resultant of two fields	B1
5(b)(iii)	$E = Q / (4\pi\epsilon_0 x^2)$	C1
	at $x = 10 \text{ cm}$, $E_A = E_B$	C1
	$Q_A / 10^2 = Q_B / 5^2$	A1
	$Q_A / Q_B = 4.0$	

37.

6(a)	work done per unit charge	B1
	(work done) moving positive charge from infinity	B1
6(b)	straight line with non-zero gradient from $x = 0$ to $x = d$	B1
	line with gradient of constant sign and end-points between which $\Delta V = V_0$ and $\Delta x = d$	B1
	line passes through $(d, 0)$ and $(0, +V_0)$ with negative gradient throughout	B1
6(c)	V constant (and non-zero) from $0 \rightarrow R$ and from $(D - R) \rightarrow D$	B1
	equal (non-zero) values of (magnitude of) V at R and $(D - R)$.	B1
	curve (with a minimum) from R to $(D - R)$ with V always positive	B1
	minimum at mid-point of curve	B1

38.

6(a)	$(E =) Q / 4\pi\epsilon_0 r^2$	M1
	where ϵ_0 is permittivity (of free space)	A1
6(b)(i)	field does not change direction/field does not become zero	M1
	so (charges have) opposite (sign)	A1
6(b)(ii)	minimum is at the midpoint (between the charges)	M1
	so (magnitudes are the) same	A1
6(c)	force = field strength $\times$ charge and force = mass $\times$ acceleration or acceleration is proportional to field strength	B1
	(from $x = 3.0$ cm) to $x = 5.0$ cm: acceleration decreases	B1
	at $x = 5.0$ cm: acceleration is a minimum	B1
	from $x = 5.0$ cm (to $x = 7.0$ cm): acceleration increases	B1

39.

9(a)	work done per unit charge	B1
	(work done) moving positive charge from infinity	B1
9(b)(i)	energy = $4.8 \times 1.60 \times 10^{-13}$ = 7.7×10^{-13} J	A1
9(b)(ii)	$E_p = Qq / 4\pi\epsilon_0 d$	C1
	$Q = 79e$ and $q = 2e$	C1
	$7.68 \times 10^{-13} = (79 \times 2 \times \{1.60 \times 10^{-19}\}^2) / (4\pi \times 8.85 \times 10^{-12} \times d)$	C1
	$d = 4.7 \times 10^{-14}$ m	A1
9(c)	(diameter must be) less than/equal to 10^{-13} or 10^{-14} m	B1

40.

6(a)	2.0 cm	B1
6(b)	At 16 (cm) from A the electric fields are equal or $E_A = E_B$	B1
	$E = Q / 4\pi\epsilon_0 r^2$	C1
	$Q_A / (4\pi\epsilon_0 r_A^2) = Q_B / (4\pi\epsilon_0 r_B^2)$	
	$3.6 \times 10^{-9} / 0.16^2 = Q_B / 0.08^2$	
	$Q_B = 9.0 \times 10^{-10}$ C	A1
6(c)(i)	$V = Q / 4\pi\epsilon_0 r_A$	C1
	$V = 3.6 \times 10^{-9} / (4 \times \pi \times 8.85 \times 10^{-12} \times 0.020)$	
	$V = 1600$ V	A1
6(c)(ii)	$C = Q / V$	C1
	$= 3.6 \times 10^{-9} / 1600$	
	$= 2.3 \times 10^{-12}$ F	A1

41.

5(a)	similarity: both are radial or both have inverse square (variations)	B1
	difference: direction is always/only towards the mass or direction can be towards or away from charge	B1
5(b)	field strength = $Q / 4\pi\epsilon_0 x^2$	C1
	$E = Q / 36\pi\epsilon_0 R^2$	A1
5(c)(i)	fields (due to each sphere) are in same direction	B1
5(c)(ii)	charges on spheres attract/affect each other or charge distribution on each sphere distorted by the other sphere or charges on the surface of the spheres move	B1
	spheres are not point charges (at their centres)	B1

42.

7(a)	force per unit charge	M1
	(force on) positive charge	A1
7(b)(i)	no electric field inside a conductor	B1
	$R = 4.5 \text{ cm}$	A1
7(b)(ii)	$E = Q / (4\pi\epsilon_0 x^2)$	C1
	clear correct read-off of a pair of values of E and x	C1
	e.g. $Q = 18 \times 10^5 \times 4\pi \times 8.85 \times 10^{-12} \times (4.5 \times 10^{-2})^2$ $= 4.0 \times 10^{-7} \text{ C}$ or $4.1 \times 10^{-7} \text{ C}$	A1
7(c)	At 8.0 cm, $E = 5.75 \times 10^5 \text{ V m}^{-1}$	C1
	$F = Eq$ and $a = F / m$	C1
	$F = (5.75 \times 10^5 \times 2 \times 1.6 \times 10^{-19}) / (4 \times 1.66 \times 10^{-27})$ $= 2.8 \times 10^{13} \text{ m s}^{-2}$	A1

43.

5(a)	work done per unit charge	B1
	(work done on charge) moving positive charge from infinity	B1
5(b)(i)	$(2.0 \times 10^{-9}) / 4\pi\epsilon_0 (4.0 \times 10^{-2}) + Q / 4\pi\epsilon_0 (8.0 \times 10^{-2}) = 0$	C1
	$Q = 4.0 \times 10^{-9} \text{ C}$	A1
	Q given with negative sign	B1
5(b)(ii)	change = 1200 V	A1
5(c)	$\frac{1}{2}mv^2 = qV$	C1
	$\frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2 = 2 \times 1.60 \times 10^{-19} \times 1200$	C1
	$v = 3.4 \times 10^5 \text{ m s}^{-1}$	A1

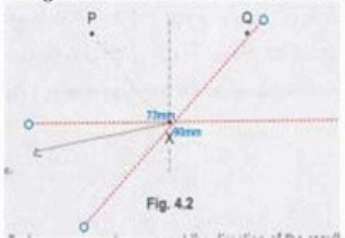
44.

5(a)(i)	region (of space)	B1
	where a particle experiences a force	B1
5(a)(ii)	similarity – any one point from: - both have an inverse square variation - both decrease with distance - both are radial	B1
	difference – any one point from: - gravitational field always towards (the mass) - electric field can be towards or away from (the charge)	B1
5(b)(i)	$E = Q / 4\pi\epsilon_0 x^2$	C1
	$Q = 4\pi \times 8.85 \times 10^{-12} \times 84 \times 0.15^2$	A1
	$= 2.1 \times 10^{-10} \text{ C}$	
5(b)(ii)	$E = 84 \times (0.15 / 0.45)^2$	C1
	or $E = (2.1 \times 10^{-10}) / (4\pi \times 8.85 \times 10^{-12} \times 0.45^2)$	
	$E = 9.3 \text{ V m}^{-1}$	A1
5(c)	line at $E = 0$ from $x = 0$ to $x = 0.15 \text{ m}$	B1
	smooth curve with decreasing negative gradient throughout, from $x = 0.15 \text{ m}$ to $x = 0.45 \text{ m}$, passing through (0.15, 84)	B1
	line passing through (0.45, 9.3)	B1

45.

6(a)	(both have) radial field lines	B1
6(b)(i)	2.1 cm	B1
6(b)(ii)	$E = \frac{Q}{4\pi\epsilon_0 r^2}$ e.g. $r = 2.1 \text{ cm}$, $E = 1.30 \times 10^5 \text{ V m}^{-1}$ $Q = 4\pi\epsilon_0 r^2 E$ $= 4 \times \pi \times 8.85 \times 10^{-12} \times 0.021^2 \times 1.30 \times 10^5$	C1
	$= 6.4 \times 10^{-9} \text{ C}$	A1
6(c)	$C = \frac{Q}{V}$ either $V = \frac{Q}{4\pi\epsilon_0 r}$ leading to $C = 4\pi\epsilon_0 r$	C1
	$C = 4 \times \pi \times 8.85 \times 10^{-12} \times 0.021$	C1
	$(C =) 2.3 \times 10^{-12} \text{ F}$	A1
	or $V = \frac{Q}{4\pi\epsilon_0 r}$ $= \frac{6.4 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.021}$ $= 2740 \text{ V}$ $C = \frac{6.4 \times 10^{-9}}{2740}$	(C1)
	$= 2.3 \times 10^{-12} \text{ F}$	(A1)

46.

4(a)	direction of force	B1
	force on a positive charge	B1
4(b)(i)	$V = \frac{Q}{4\pi\epsilon_0 r}$ $\frac{4.0 \times 10^{-9}}{4\pi\epsilon_0 x} + \frac{-7.2 \times 10^{-9}}{4\pi\epsilon_0 (0.120 - x)} = 0$ $4(0.120 - x) = 7.2 x$ $x = 0.043 \text{ m}$	C1
		A1
4(b)(ii)	fields are in the same direction so no	B1
4(b)(iii)	straight arrow drawn leftwards from X in direction between extended line joining Q and X and the horizontal 	B1

47.

2(a)(i)	(vertically) downwards	B1
2(a)(ii)	magnetic force (on sphere) is perpendicular to its velocity	B1
	magnetic force perpendicular to velocity is the centripetal force or magnetic force perpendicular to velocity causes centripetal acceleration or acceleration perpendicular to velocity is centripetal (acceleration) or magnetic force does not change the speed of the sphere or magnetic force has constant magnitude	B1
2(b)	$mg = Eq$	C1
	$E = (1.6 \times 10^{-10} \times 9.81) / (0.27 \times 10^{-9})$ $= 5.8 \text{ N C}^{-1}$	A1
2(c)	centripetal force = magnetic force or $Bqv = mv^2 / r$	B1
	$B = mv / qr$	C1
	$= (1.6 \times 10^{-10} \times 0.78) / (0.27 \times 10^{-9} \times 3.4) = 0.14 \text{ T}$	A1

48.

2(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
2(b)(i)	(electric) force is perpendicular to velocity (of particles)	B1
	force (perpendicular to velocity) causes centripetal acceleration or force does not change the speed of the particles or force has constant magnitude	B1
2(b)(ii)	$F = e^2 / 4\pi\epsilon_0 x^2$	C1
	$= (1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (2 \times 1.59 \times 10^{-10})^2]$ $= 2.28 \times 10^{-9} \text{ N}$	A1
2(b)(iii)	$F = m r \omega^2$ and $\omega = 2\pi / T$ or $F = m v^2 / r$ and $v = 2\pi r / T$	C1
	$F = 4\pi^2 m r / T^2$ $T = \sqrt{[4\pi^2 \times 9.11 \times 10^{-31} \times 1.59 \times 10^{-10} / (2.28 \times 10^{-9})]}$ $= 1.58 \times 10^{-15} \text{ s}$	C1
2(c)(i)	- electron and positron interact - positron is anti-particle of electron (pair) annihilation occurs <i>Any two points, 1 mark each</i>	B2
	mass of the electron and positron converted into photon energy	B1
2(c)(ii)	PET scanning	B1

49.

4(a)	(field line indicates) direction of force	B1
	force on a positive charge	B1
4(b)(i)	one straight line perpendicular to plates, starting on one plate and finishing on the other	B1
	five straight lines perpendicular to plates between the plates, uniformly spaced	B1
	downwards arrows on lines	B1
4(b)(ii)	$E = V / d$	C1
	$= 2400 / 0.046$ $= 5.2 \times 10^4 \text{ N C}^{-1}$	A1
4(c)(i)	smooth curve in region of field and straight line outside field	B1
	direction of deflection shown as downwards in region of field	B1
4(c)(ii)	helium nucleus has double the charge but four times the mass	B1
	velocity parallel to plates same and acceleration perpendicular to plates smaller (for helium)	B1
	final speed is lower (for helium)	B1

50.

5(a)	work done per unit charge	B1
	work done (on charge) in moving positive charge from infinity (to the point)	B1
5(b)(i)	radius = 0.060 m	A1
5(b)(ii)	$V = Q / 4\pi\epsilon_0 x$ $Q = (-) 850 \times 4\pi \times 8.85 \times 10^{-12} \times 0.060$ or $Q = (-) 850 \times 0.060 / 8.99 \times 10^9$ <i>(any correct pair of V and x values from curve)</i>	C1
	$Q = -5.7 \times 10^{-9} \text{ C}$	A1
5(c)(i)	$E_P = Q^2 / 4\pi\epsilon_0 x$ $= (5.67 \times 10^{-9})^2 / (4\pi \times 8.85 \times 10^{-12} \times 0.46)$	C1
	$= 6.3 \times 10^{-7} \text{ J}$	A1
5(c)(ii)	- force is repulsive so spheres move apart - force in direction of motion so speed increases - potential energy converted to kinetic energy so speed increases - force decreases with distance so acceleration decreases - momentum is conserved (at zero) (and masses are equal) so velocities are always equal and opposite <i>Any three points, 1 mark each</i>	B3

51.

4(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
4(b)(i)	arrows showing tension upwards in direction of string, electric force horizontally to the right and weight vertically downwards and all three labelled	B1
4(b)(ii)	$F_E = \frac{96 \times 10^{-9} \times 64 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.080^2}$ $(= 8.63 \times 10^{-3} \text{ N})$	C1
	either angle to vertical $= \sin^{-1} 0.080 / 1.2$ $(= 3.82^\circ)$	C1
	weight $= F_E / \tan 3.82 = 8.63 \times 10^{-3} / \tan 3.82$ $(= 0.129 \text{ N})$	C1
	mass $= 0.129 / 9.81$ $= 0.013 \text{ kg}$	A1
	or $T \sin \theta = mg$ and $T \cos \theta = F_E$ or $\tan \theta = mg / F_E$	(C1)
	$\tan \theta = 1.2 / 0.080$	(C1)
	$m = (1.2 \times 8.63 \times 10^{-3}) / (0.080 \times 9.81)$ $= 0.013 \text{ kg}$	(A1)

4(b)(iii)	$E_p = \frac{Q_1 Q_2}{4\pi\epsilon_0 r} = \frac{96 \times 10^{-9} \times 64 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.080}$ $= 6.9 \times 10^{-4} \text{ J}$	A1
4(c)(i)	towards the top of the page / towards plate P	B1
4(c)(ii)	$F = QE \text{ and } E = V/d$ $F = 1.6 \times 10^{-19} \times 250/0.018$ $= 2.2 \times 10^{-15} \text{ N}$	C1 A1
4(c)(iii)	either the force is not (always) perpendicular to the velocity or the force is always in the same direction	B1

52.

1(a)(i)	force per unit mass	B1
1(a)(ii)	force per unit positive charge	B1
1(a)(iii)	similarity: - inversely proportional to distance (from point) - points of equal potential lie on concentric spheres - zero at infinite distance Any point, 1 mark	B1
	difference: - gravitational potential is (always) negative - electric potential can be positive or negative Any point, 1 mark	B1
1(b)(i)	$g = GM/r^2$	M1
	$E = Q/4\pi\epsilon_0 r^2$	M1
	algebra showing the elimination of r leading to $M/Q = (1/4\pi G\epsilon_0)(g/E)$	A1
1(b)(ii)	$\alpha = 1/(4\pi \times 6.67 \times 10^{-11} \times 8.85 \times 10^{-12}) = 1.35 \times 10^{20} \text{ (kg}^2 \text{ C}^{-2}\text{)}$ or $\alpha = (8.99 \times 10^9)/(6.67 \times 10^{-11}) = 1.35 \times 10^{20} \text{ (kg}^2 \text{ C}^{-2}\text{)}$	A1
1(c)(i)	$E = \alpha g Q/M$ $= (1.35 \times 10^{20} \times 9.81 \times 4.80 \times 10^5)/(5.98 \times 10^{24})$ $= 106 \text{ N C}^{-1} \text{ or } 106 \text{ V m}^{-1}$	C1 A1
1(c)(ii)	same (direction)	B1

53.

5(a)	work done per unit charge	B1
	work (done on charge) moving positive charge from infinity (to the point)	B1
5(b)	Any three points from: Up to 2 points from: - radius of sphere X is 2.0 m - radius of sphere Y is 4.0 m - radius of Y is double the radius of X Up to 2 points from: - charge on X is negative - charge on Y is positive - spheres carry opposite charges Up to 1 point from: - magnitudes of charges on the spheres are equal	B3
5(c)	particle is attracted to X or repelled from Y or resultant force on particle is towards X / away from Y / to the left	B1
	particle accelerates towards X / away from Y / to the left	B1
	(magnitude of) acceleration of particle increases	B1

54.

5(a)	(electric) force is (directly) proportional to product of charges	B1
	(electric) force (between point charges) is inversely proportional to the square of their separation	B1
5(b)	$F = Q^2 / 4\pi\epsilon_0 x^2$ $6.3 \times 10^{-17} = Q^2 / [4\pi \times 8.85 \times 10^{-12} \times (3.8 \times 10^{-6})^2]$	C1
	charge = 3.2×10^{-19} C	A1
5(c)(i)	negative	B1
5(c)(ii)	four straight lines perpendicular to the plates, starting on one plate and finishing on the other	B1
	lines equally spaced	B1
	arrows indicating direction downwards	B1
5(c)(iii)	$E = V / d$	C1
	$mg = EQ$	C1
	mass = $(1200 \times 3.2 \times 10^{-19}) / (9.81 \times 0.052)$ = 7.5×10^{-16} kg	A1